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Inventor(s)	Abunassar; Chad

Catheter assembly with coaptation aid and methods for valve repair

Abstract

Catheter assembly with a coaptation aid for repair of leaflets of a heart valve includes an elongate catheter shaft adapted to be positioned transvascularly proximate a heart valve and a lever coupled to the catheter shaft. The lever has a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with the catheter shaft and a deployed configuration extending transversely from the catheter shaft. The lever also includes an atraumatic tip at a distal end thereof and adapted to contact a distal side of a native leaflet of the heart valve when the catheter shaft is positioned proximate the heart valve and the lever is in the deployed configuration, and an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration. Method for fixation of native leaflets of a heart valve using the catheter assembly also disclosed.

Inventors:	Abunassar; Chad (Alameda, CA)
Applicant:	Evalve, Inc. (Santa Clara, CA)
Family ID:	73740559
Assignee:	Evalve, Inc. (Santa Clara, CA)
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Primary Examiner: Aleman; Sarah W

Attorney, Agent or Firm: SLEMAN & LUND LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/096,638, filed on Nov. 12, 2020, which claims the benefit of U.S. Provisional Application No. 62/935,227, filed Nov. 14, 2019, the disclosures of which are incorporated herein by reference in their entirety.

FIELD OF DISCLOSED SUBJECT MATTER

(1) The disclosed subject matter is directed to medical devices and methods for the endovascular, percutaneous or minimally invasive surgical treatment of bodily tissues, such as tissue approximation or valve repair. More particularly, the present disclosure relates to repair of valves of the heart and venous valves.

(2) Surgical repair of bodily tissues can involve tissue approximation and fastening of such tissues in the approximated arrangement. When repairing valves, tissue approximation includes coapting the leaflets of the valves in a therapeutic arrangement, which can then be maintained by fastening or fixing the leaflets. Such coaptation can be used to treat regurgitation, which commonly occurs in the mitral valve and in the tricuspid valve.

(3) Mitral valve regurgitation is characterized by retrograde flow from the left ventricle of a heart through an incompetent mitral valve into the left atrium. During a normal cycle of heart contraction (systole), the mitral valve acts as a check valve to prevent flow of oxygenated blood back into the left atrium. In this way, the oxygenated blood is pumped into the aorta through the aortic valve. Regurgitation of the mitral valve can significantly decrease the pumping efficiency of the heart, placing the patient at risk of severe, progressive heart failure.

(4) Mitral valve regurgitation can result from a number of different mechanical defects in the mitral valve or the left ventricular wall. The valve leaflets, the valve chordae, which connect the leaflets to the papillary muscles, the papillary muscles or the left ventricular wall can be damaged or otherwise dysfunctional. Commonly, the valve annulus can be damaged, dilated, or weakened, limiting the ability of the mitral valve to close adequately against the high pressures of the left ventricle. In some anatomies, the anterior mitral leaflet can be positioned at a significant distance away from the opposing posterior mitral leaflet or the anterior leaflet can dynamically move away from the region of coaptation with the posterior leaflet, and there is a need for repair assemblies enabling leaflet grasping in such or similar anatomies.

SUMMARY

(5) The purpose and advantages of the disclosed subject matter will be set forth in and apparent from the description that follows, as well as will be learned by practice of the disclosed subject matter. Additional advantages of the disclosed subject matter will be realized and attained by the methods and systems particularly pointed out in the written description and claims hereof, as well as from the appended drawings.

(6) To achieve these and other advantages and in accordance with the purpose of the disclosed subject matter, as embodied and broadly described, the disclosed subject matter is directed to a catheter assembly with a coaptation aid for repair of leaflets of a heart valve. The catheter assembly includes an elongate catheter shaft having a proximal portion and a distal portion adapted to be positioned transvascularily proximate a heart valve. The catheter assembly includes a lever coupled to the distal portion of the catheter shaft, the lever having a proximal end, a distal end, and a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with the distal portion of the catheter shaft and a deployed configuration extending transversely from the distal portion of the catheter shaft. The lever also has an atraumatic tip proximate the distal end thereof adapted to contact a distal side of a native leaflet of the heart valve when the distal portion of the catheter shaft is positioned proximate the heart valve and the lever is in the deployed configuration. The catheter assembly also includes an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration.

(7) The actuation assembly can include a transmission member extending along a length of the catheter shaft to remotely transition the lever between the delivery configuration and the deployed configuration. The transmission member can include a pull wire to transition the lever from the delivery configuration to the deployed configuration by rotating the lever about the hinge. The lever can be biased toward the delivery configuration. The pull wire can extend through an alignment loop disposed on the distal portion of the catheter shaft. The pull wire can extend through a lumen along a length of the catheter shaft. The actuation assembly can further include a linkage coupled to the lever with the pull wire attached to the linkage. The linkage can be coupled to the lever between the hinge and the distal end.

(8) The atraumatic tip can comprise rounded edges and can comprise one or more of nitinol, cobalt chromium, stainless steel, titanium, polyether block amide (PEBAX), polylactic acid (PLA), poly-L-lactide (PLLA), polylactic-co-glycolic acid (PLGA), polyurethane, polyethylene, polyester, and polyamide.

(9) The catheter assembly can further include a second lever coupled to the distal portion of the catheter shaft, the second lever having a proximal end, a distal end, and a hinge to transition the

second lever between the delivery configuration and the deployed configuration. The second lever can further include a second atraumatic tip proximate the distal end of the second lever adapted to contact the distal side of the native leaflet. The actuation assembly can include a transmission member extending along a length of the catheter shaft to remotely transition the lever and the second lever simultaneously between the delivery configuration and the deployed configuration. The transmission member can include a pull wire attached to a linkage coupled to both the lever and the second lever to simultaneously transition the lever and the second lever from the delivery configuration to the deployed configuration by rotating each lever about its hinge. The pull wire can extend through an alignment loop disposed on the distal portion of the catheter shaft. The pull wire can extend through a lumen along a length of the catheter shaft.

(10) The hinge can be defined as a preset bend in the lever. The actuation assembly can include a slidable collar disposed on the lever. The collar can be slidable between a delivery condition positioned over the hinge to straighten the bend of the lever to the delivery configuration and a deployed condition with the bend exposed to allow the lever to transition to the deployed configuration. The lever can be made of a super elastic material. The actuation assembly can include a transmission member, which can be a rod adapted to slide the collar between the delivery condition and the deployed condition. The rod can be a hypotube.

(11) The lever can include a varied mechanical property along a length thereof. The varied mechanical property can be selected from a varied thickness or a varied material property. The varied mechanical property can include a stiffening member along a length thereof.

(12) The lever can be made of a material selected from the group consisting of a metal, a polymer, a composite, or a combination thereof. The lever can include a wire.

(13) The catheter assembly can further include a leaflet fixation device releasably coupled to the distal portion of the catheter shaft. The leaflet fixation device can include a first gripper assembly and a second gripper assembly, each gripper assembly configured to capture a respective native leaflet.

(14) In accordance with the disclosed subject matter, a method for fixation of native leaflets of a heart valve includes introducing a catheter assembly transvascularly, the catheter assembly having an elongate catheter shaft with a proximal portion and a distal portion. The catheter assembly also includes a lever coupled to the distal portion of the catheter shaft. The lever has a proximal end, a distal end, and a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with the distal portion of the catheter shaft and a deployed configuration extending transversely from the distal portion of the catheter shaft. The lever also includes an atraumatic tip proximate the distal end thereof. The catheter assembly also includes an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration. The catheter assembly also includes a leaflet fixation device releasably coupled to the distal portion of the catheter shaft and including a first gripper assembly and a second gripper assembly. The method also includes positioning the catheter with the distal portion positioned proximate the heart valve and deploying the first gripper assembly to capture a first native leaflet, actuating the actuation assembly to remotely transition the lever from the delivery configuration toward the deployed configuration to position the atraumatic tip against the distal side of a second native leaflet, and deploying the second gripper assembly to capture the second native leaflet.

(15) The method for fixation of native leaflets of a heart valve can include any of the features described above for the catheter assembly with a coaptation aid.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1A is a trans-esophageal echocardiography image of a mitral valve; FIG. 1B is a schematic illustration of the mitral valve anatomy of FIG. 1A; and FIG. 1C is a schematic illustration of the mitral valve anatomy of FIG. 1A having a leaflet fixation device for edge-to-edge repair positioned to reach both leaflets of the mitral valve.

(2) FIG. 2A is a schematic illustration of mitral valve anatomy having a depressed anterior mitral leaflet; FIG. 2B is a schematic illustration of the mitral valve anatomy of FIG. 2A having a leaflet fixation device for edge-to-edge repair positioned proximate the posterior leaflet and not capable of reaching the anterior mitral leaflet.

(3) FIGS. 3A, 3B, and 3C are front views of an exemplary embodiment of a catheter assembly with a coaptation aid for repair of leaflets of a heart valve and illustrate a sequential schematic illustration of a method for fixation of native leaflets of a heart valve in accordance with the disclosed subject matter.

(4) FIGS. 4A and 4B are front views of another exemplary embodiment of a catheter assembly with coaptation aid for repair of leaflets of a heart valve having two levers in accordance with the disclosed subject matter.

(5) FIG. 5 is a front view of another exemplary embodiment of a catheter assembly with coaptation aid for repair of leaflets of a heart valve having an alignment loop in accordance with the disclosed subject matter.

(6) FIGS. 6A and 6B are front views of another exemplary embodiment of a catheter assembly with coaptation aid for repair of leaflets of a heart valve having a slidable collar and illustrate a method for fixation of native leaflets of a heart valve in accordance with the disclosed subject matter.

(7) FIGS. 7A and 7B are front views of another exemplary embodiment of a catheter assembly with coaptation aid for repair of leaflets of a heart valve having a slidable collar in accordance with the disclosed subject matter.

(8) FIGS. 8A and 8B are front views of another exemplary embodiment of a catheter assembly with coaptation aid for repair of leaflets of a heart valve having a slidable collar in accordance with the disclosed subject matter.

(9) FIGS. 9A, 9B, 9C, 9D, 9E and 9F are front views of various exemplary embodiments of an atraumatic tip for a coaptation aid for repair of leaflets of a heart valve in accordance with the disclosed subject matter.

DETAILED DESCRIPTION

(10) Reference will now be made in detail to various exemplary embodiments of the disclosed subject matter, which are illustrated in the accompanying drawings. A technique for mitral valve repair, which relies on suturing adjacent segments of the opposed valve leaflets together is referred to as the “bow-tie” or “edge-to-edge” technique. The catheter assemblies and methods of the disclosed subject matter provide for repair of leaflets of a heart valve, such as edge-to-edge valve repair for patients having various conditions, such as a regurgitant mitral valve. Such assemblies and method likewise can be useful for repair of tissues in the body other than heart valves. The assemblies and methods disclosed herein do not require open chest access and are capable of being performed endovascularly, i.e., using devices, such as a catheter, which are advanced to the heart from a point in the patient's vasculature remote from the heart.

(11) For the purpose of illustration and not limitation, FIG. 1A is trans-esophageal echocardiography image of a heart valve **100**, specifically a left ventricular outflow tract view of a mitral valve. Such a trans-esophageal echocardiography can be used for visualization during an edge-to-edge mitral valve repair procedure.

(12) FIG. 1B is a schematic illustration of the mitral valve **100** of FIG. 1A and depicts the mitral anatomy including an anterior mitral leaflet **103** and a posterior mitral leaflet **105** that open and close to control blood flow from the left atrium **101** to the left ventricle **102** and further through the

left ventricular outflow tract **106**. FIG. 1B (and the trans-esophageal echocardiography of FIG. 1A) shows a mild regurgitation disease state, where the mid-sections of the anterior mitral leaflet **103** (A2 in FIG. 1A) and posterior mitral leaflet **105** (P2 in FIG. 1A) are shown to be non-coapting, but to be reasonably close to each other during cardiac systole. Edge-to-edge valve repair can be performed using a leaflet fixation device by targeting each leaflet to improve coaptation of the anterior and posterior mitral leaflets and reduce regurgitation.

(13) As illustrated in FIG. 1C, when performing transcatheter edge-to-edge repair, a leaflet fixation device **150** is delivered on the distal end of a catheter shaft **111** such that the opposing leaflets **103**, **105** are inserted within a leaflet fixation device **150** and brought together into coaptation. These fixation devices can use arms to contact the distal side **104** of leaflets. Once the arms are positioned to stabilize the leaflets in a beating heart, gripping elements or the like contact the atrial side of the leaflets to capture the leaflets. Once both opposing leaflets are captured, the fixation device **150** can be closed such that the leaflets are pulled and brought into coaptation, which results in a reduction in valvular regurgitation during ventricular systole.

(14) However, leaflet locations can vary from patient to patient, and in certain patients it can be difficult to reach and capture both leaflets using fixation devices alone. For example, FIG. 2A depicts a mitral valve anatomy demonstrating a large coaptation gap, with the anterior mitral leaflet **103** depressed or spaced at a distance from the posterior mitral leaflet **105**. As illustrated in FIG. 2B, because the anterior mitral leaflet **103** is significantly spaced from the posterior mitral leaflet **105**, the leaflet fixation device **150** may not be able to reach the anterior mitral leaflet **103**, and successful edge-to-edge repair may be difficult, requiring a great deal of procedural skill and time. The leaflet grasping procedure can be further complicated by erratic leaflet motion, such as systolic anterior motion of the anterior mitral valve leaflet. Further, if only a minimal amount of leaflet can be inserted within the arms and gripping elements during the procedure based upon the large gap, mitral regurgitation may not be significantly improved by a single implantation and additional devices or further intervention may be needed to reduce the mitral regurgitation.

(15) To address these problems, generally, and as set forth in greater detail below, the disclosed subject matter includes a catheter assembly with a coaptation aid for repair of leaflets of a heart valve. The catheter assembly includes an elongate catheter shaft having a proximal portion and a distal portion adapted to be positioned transvascularly proximate a heart valve. The catheter assembly includes a lever coupled to the distal portion of the catheter shaft, the lever having a proximal end, a distal end, and a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with the distal portion of the catheter shaft and a deployed configuration extending transversely from the distal portion of the catheter shaft. The lever also has an atraumatic tip proximate the distal end thereof adapted to contact a distal side of a native leaflet of the heart valve when the distal portion of the catheter shaft is positioned proximate the heart valve and the lever is in the deployed configuration. The catheter assembly also includes an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration.

(16) Likewise, as further described in conjunction with the device of the disclosed subject matter, a method for fixation of native leaflets of a heart valve is provided. The method includes introducing a catheter assembly transvascularly, the catheter assembly having an elongate catheter shaft with a proximal portion and a distal portion. The catheter assembly also includes a lever coupled to the distal portion of the catheter shaft. The lever has a proximal end, a distal end, and a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with the distal portion of the catheter shaft and a deployed configuration extending transversely from the distal portion of the catheter shaft. The lever also includes an atraumatic tip proximate the distal end thereof. The catheter assembly also includes an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration. The catheter assembly also includes a leaflet fixation device releasably coupled to the distal portion

of the catheter shaft and including a first gripper assembly and a second gripper assembly. The method also includes positioning the catheter with the distal portion positioned proximate the heart valve and deploying the first gripper assembly to capture a first native leaflet, actuating the actuation assembly to remotely transition the lever from the delivery configuration toward the deployed configuration to position the atraumatic tip against the distal side of a second native leaflet, and deploying the second gripper assembly to capture the second native leaflet.

(17) The coaptation aid thus can improve the ease of leaflet grasping and capture by providing a mechanism to temporarily facilitate positioning and stabilization of a leaflet, such as the anterior mitral leaflet, during an edge-to-edge repair procedure.

(18) Referring to FIG. 3A, for purpose of illustration and not limitation, an exemplary catheter assembly **300** includes an elongate catheter shaft **111** having a proximal portion **312** and a distal portion **313**. The catheter assembly **300** includes a coaptation aid **320** having a lever **321** coupled to the distal portion **313** of the catheter shaft. The lever **321** includes a proximal end **322**, a distal end **323**, and a hinge **324**. An actuation assembly **340** is coupled to the lever **321** to remotely transition the lever. The lever **321** has an atraumatic tip **325** proximate the distal end **323** thereof. A leaflet fixation device **150**, e.g., a clip, is releasably coupled to the distal portion **313** of the catheter shaft and is configured to grasp and couple two (or more) leaflets together. Although a number of edge-to-edge leaflet repair devices are known, for purpose of understanding and not limitation, reference will be made herein to a fixation device having a first gripper assembly **351** and a second gripper assembly **352** to capture the native leaflets, wherein each gripper assembly includes a movable arm and gripper element to capture a native leaflet therebetween.

(19) In use, the catheter assembly **300** is introduced proximate the native leaflets of a heart valve **100**, for example using an antegrade approach from a patient's left atrium **101** to the left ventricle **102**. In a transcatheter approach, the catheter assembly **300** can be introduced in a femoral vein and advanced through the inferior vena cava into the heart, across a penetration in the interatrial septum (i.e., a transseptal approach), and to the mitral valve from the atrium toward the ventricle. Details of various suitable approaches for leaflet repair are set forth, for example, in U.S. Pat. No. 7,226,467 to Lucatero et al., U.S. Pat. No. 7,563,267 to Goldfarb et al., U.S. Pat. No. 7,655,015 to Goldfarb et al., U.S. Pat. No. 7,736,388 to Goldfarb et al., U.S. Pat. No. 7,811,296 to Goldfarb et al., U.S. Pat. No. 8,057,493 to Goldfarb et al., U.S. Pat. No. 8,303,608 to Goldfarb et al., U.S. Pat. No. 8,500,761 to Goldfarb et al., U.S. Pat. No. 8,734,505 to Goldfarb et al., U.S. Pat. No. 8,740,920 to Goldfarb et al., U.S. Pat. No. 9,510,829 to Goldfarb et al., U.S. Pat. No. 7,635,329 to Goldfarb et al., U.S. Patent Application Publication No. 2017/0042546 to Goldfarb et al., U.S. Patent Application Publication No. 2017/0239048 to Goldfarb et al., U.S. Patent Application Publication No. 2018/0325671 to Abunassar et al., the contents of each of which is incorporated by reference in its entirety herein (collectively “the Representative Patent Publications”).

(20) Once located proximate native leaflets of a heart valve **100**, the first gripper assembly **351** of the leaflet fixation device **150** is deployed to capture a first native leaflet, such as the posterior mitral leaflet **105**, as depicted in FIG. 3A. For example, arms of the fixation device **150** can be aligned generally perpendicular to a line of coaptation of the mitral valve, with the device positioned so that a first arm contacts the ventricular surface of the posterior mitral leaflet **105**. A corresponding gripping element can remain on the atrial side of the valve leaflet so that the leaflet lies between the gripping element and the arm. As embodied herein, the first gripping element can be lowered toward the arm so that the leaflet is captured and held therebetween. Alternatively, the leaflet can be held with the arm underneath without lowering the gripping element until later in the procedure. Additional features and alternative embodiments of leaflet fixation devices and the grasping and capture process are described in more detail in the Representative Patent Publications.

(21) In certain anatomies where the anterior mitral leaflet is significantly spaced from the posterior mitral leaflet or has dynamically moved away from the region of coaptation where edge-to-edge repair using the leaflet fixation device would not be possible, the coaptation aid of the disclosed

subject matter can be used to position the anterior mitral leaflet close to the posterior mitral leaflet to allow for edge-to-edge repair.

(22) For example, lever **321** of the coaptation aid **320** is transitioned to locate the atraumatic tip **325** on the distal side **104** of the leaflet as shown in FIG. **3B** and described in more detail below. The lever **321** can be made of a material selected from the group consisting of a metal, a polymer, a composite, or a combination thereof. For example, the lever **321** can include a wire. The tip **325** can be configured to be atraumatic, e.g., by using a suitable construction (as further detailed herein below) such as a flexible polymeric solid or hollow component, a flexible nitinol mesh component, or a flexible metallic component. A metallic version of the tip can be coated or jacketed with polymeric material such as PEBAX. Exemplary materials that can be used for the atraumatic tip include any combination of nitinol, cobalt chromium, stainless steel, titanium, polyether block amide (PEBAX), polylactic acid (PLA), poly-L-lactide (PLLA), polylactic-co-glycolic acid (PLGA), polyurethane, polyethylene, polyester, polyamide, and other suitable materials. The tip **325** can also be configured to be atraumatic, e.g., by using rounded edges and/or a tapered configuration, to reduce damage to the leaflets, chordae tendinae, papillary muscles, ventricular wall, and other heart tissue. Additionally, the atraumatic tip **325** should be dimensioned sufficiently slender to reduce entanglement with subvalvular structures including the papillary muscles and/or chordae tendinae when navigating through these structures.

(23) With continued reference to FIG. **3A**, the lever **321** has elongate configuration **330** during delivery to the heart valve **100**, in which the lever is generally aligned longitudinally with and extends beyond the distal portion **313** of the catheter shaft to minimize profile and the bending stiffness of the catheter system. As embodied herein, the lever **321** is transitioned from the delivery configuration **330** via rotation about hinge **324** such that atraumatic tip **325** navigates through chordae tendinae and contacts a distal side **104** of the leaflet and begins to push the anterior leaflet **103** inward, as shown in a partially rotated configuration shown in FIG. **3B**.

(24) For example, and in accordance with the disclosed subject matter, in order to transition the lever **321** between the delivery configuration **330**, as depicted schematically in FIG. **3A**, and the deployed configuration **331**, as depicted schematically in FIG. **3C**, the catheter assembly **300** can include an actuation assembly **340** coupled to the lever **321**. The actuation assembly **340** can include a transmission member **341**, such as a wire, suture, or any suitable means, extending along a length of the catheter shaft **111**. For example, the transmission member **341** can include a pull wire **342**. The pull wire can extend through an alignment loop **344** disposed on the distal portion **313** of the catheter shaft to help guide the pull wire **342** toward a proximal portion (e.g., handle) of the catheter assembly (not shown) and reduce the chance of engagement with the catheter assembly **300** or cardiac structures. Additionally or alternatively, the pull wire can extend through a lumen along a length of the catheter shaft. Additional features and alternative embodiments of suitable catheter assemblies (including details of the proximal portion) are described in more detail in the Representative Patent Publications.

(25) As embodied herein, the lever **321** can be biased toward the delivery configuration **330** and can be transitioned by applying tension to the pull wire **342** to transition the lever **321** from the delivery configuration **330** (FIG. **3A**) to the deployed configuration **331** (FIG. **3C**) by rotating the lever about the hinge **324** to position the atraumatic tip **325** against the distal side **104** of the native leaflet.

(26) As embodied herein, with reference again to FIG. **3A**, the actuation assembly **340** can further include a linkage **343** coupled to the lever **321**. The linkage can include a wire rotatably connected to the lever **321** via a rivet or other suitable connecting means proximate a first end. The pull wire **342** can be attached to the linkage **343** proximate a second end. The linkage can be coupled to the lever **321** between the hinge **324** and the distal end **323** of the lever at a pivot distance to form a fulcrum and enable rotation of the lever about the hinge point.

(27) The force applied to the leaflet to pull it into coaptation is proportional to the tension applied

on the pull wire **342** and depends on the distance between the linkage **343** attachment location and the hinge **324** point and the distance between the atraumatic tip **325** and the hinge **324** point. Suitable distances between the linkage **343** attachment location and the hinge **324** point is about 5-10 mm and between the atraumatic tip **325** and the hinge **324** point is about 15-20 mm. In use, the coaptation load should not exceed about 0.8 to 1.0 lbf for an extended amount of time to reduce damage to leaflets and other cardiac tissue. Thus, the lever **321** can be adapted to apply a force of less than about 1.0 lbf to the native leaflet when the lever **321** is transitioned to the deployed configuration **331**. The geometry shown in FIGS. **3A**, **3B** and **3C** is exemplary, and one of ordinary skill in the art will recognize that any relative linkage attachment location, hinge point, curvature, damper or break in the hinge, variable thickness features, or stiffening features may be adjusted or additionally used to ensure loads do not impart excessive stress on the leaflets or the coaptation aid structure or its attachments.

(28) For purpose of illustration, FIG. **3B** depicts a modest amount of pull wire **342** tension being applied, which rotates the coaptation aid **320** such that the atraumatic tip **325** is brought into contact with the distal side **104** of the anterior leaflet **103**. FIG. **3C** shows the effect of further tension being applied on the pull wire **342**, which swings the lever **321** and atraumatic tip **325** upward until a fully deployed configuration **331** is achieved. This rotation applies tension to the anterior leaflet **103** to bring it into a position located sufficiently close to the posterior leaflet **105** such that both leaflets can be grasped and captured by arms of the leaflet fixation device **150**. In some anatomies, as depicted in FIG. **3B**, in the deployed configuration **331**, the lever **321** extends generally transversely from the distal portion **313** of the catheter shaft. Alternatively, in some anatomies, the lever **321** can fully rotate and extend generally aligned longitudinally with the distal portion **313** of the catheter shaft as depicted in FIG. **3C**.

(29) In the deployed configuration **331** of FIG. **3C**, with the coaptation aid **320** acting as a stabilizing support for the anterior mitral leaflet **103**, the fixation device **150** can be more easily used to simultaneously grasp both leaflets. The arm of the second gripper assembly **352** of the leaflet fixation device **150** is deployed to contact the ventricular surface **104** of and grasp the anterior mitral leaflet **103**. The corresponding gripping element can remain on the atrial side of the leaflet so that the leaflet lies between the gripping element and the arm.

(30) If needed, before the second gripper is deployed, the fixation device **150** can be repeatedly manipulated to reposition the device so that the leaflets are properly grasped at a desired location. Repositioning can be achieved with the fixation device **150** in an open position. For example, the coaptation aid **320** can be manipulated to reposition the leaflets, such as by applying additional tension or reducing tension on the pull wire **342** to better position the anterior mitral leaflet **103**. As embodied herein, the catheter shaft **111** and or lever **321** of the coaptation aid can be longitudinally rotated to apply different tension on the leaflet via the atraumatic tip **325** to better position the leaflet for capture. Regurgitation of the valve can also be checked while the fixation device **150** is in the open position. If regurgitation is not satisfactorily reduced, the fixation device **150** can be repositioned, and regurgitation checked again until desired results are achieved.

(31) Once the fixation device **150** has been positioned in a desired location relative to the valve leaflets, the leaflets can then be captured between the gripping elements and the arms. As noted above, the first gripping element may already have been lowered to capture the posterior leaflet. If so, the second gripping element is lowered to capture the anterior leaflet. Alternatively, if the posterior leaflet has not yet been captured, both gripping elements can be lowered toward the arms so that the leaflets are held and captured therebetween, either simultaneously or sequentially. Once both leaflets are captured, the arms can be closed to a tight arm angle to bring each leaflet into central coaptation to reduce regurgitation. The arms can further be locked to prevent the device from moving toward an open position.

(32) Once completed, the coaptation aid **320** can be removed by releasing tension on the pull wire **342**. As tension is released, the coaptation aid **320** will not have any bending stiffness and will

align with the delivery catheter for safe withdrawal from the body. The repair of the leaflets or tissue can be observed by non-invasive visualization techniques, such as echocardiography, to ensure the desired outcome. If the repair is not desired, the fixation device **150** can be repositioned or retrieved. Once the desired outcome is achieved, the fixation device **150** can be detached from the distal portion **313** of the catheter shaft, and the catheter assembly **300** with the coaptation aid **320** can be withdrawn from the body.

(33) In accordance with another aspect of the disclosed subject matter, referring now to FIGS. **4A** and **4B**, the catheter assembly **300** can further include a second lever **461** coupled to the distal portion **313** of the catheter shaft, the second lever having a proximal end **462**, a distal end **463**, and a hinge **464** to transition the second lever between the delivery configuration **330** and the deployed configuration **331**. For example, the second lever **461** can be coupled to the catheter shaft **111** on a side opposite the first lever **321**. The second lever **461** can also include a second atraumatic tip **465** proximate the distal end **463** thereof adapted to contact the distal side **104** of the native leaflet. The hinge **464** can be a common hinge mechanism so that the two separate atraumatic tips **325**, **465**, rotate about the common hinge.

(34) In embodiments with two levers and with continued reference to FIG. **4A**, the actuation assembly **340** can include a transmission member **341** extending along a length of the catheter shaft **111**. The transmission member **341** can include a wire, suture, or other suitable means, to remotely transition the lever **321** and the second lever **461** simultaneously between the delivery configuration and the deployed configuration. For example, the transmission member **341** includes a pull wire **342** attached to a linkage **343** coupled to both the lever **321** and the second lever **461** to simultaneously transition the lever and the second lever from the delivery configuration to the deployed configuration by rotating each lever about its hinge to position each atraumatic tip **325**, **465** against the distal side **104** of the native leaflet.

(35) With reference now to FIG. **5**, as embodied herein, the pull wire **342** can extend through an alignment loop **544** disposed on the distal portion **313** of the catheter shaft. The alignment loop **544** can be a coaxial tether loop in order to route any pull wire **342** slack along a central axis of the catheter assembly **300** toward a proximal portion (e.g., handle) of the catheter assembly (not shown). The alignment loop **544** can be any suitable configuration to reduce entanglement or inference with the fixation device arms closing together on opposing leaflets. Additionally or alternatively, the pull wire can extend through a lumen along a length of the catheter shaft.

(36) Embodiments of the disclosed subject matter with two levers can be used for valve repair, for example in edge-to-edge repair, in the same manner as described herein above. As embodied herein, each of the two atraumatic tips **325**, **465** can be positioned just adjacent to the fixation device arms at a width dimension between, such as 6 mm to 10 mm, so that the leaflet areas on either side of the fixation device arm are supported during the repair procedure. Thus, the location of these tips ensures that the closing of the fixation devices arms is not obstructed.

(37) In accordance with another aspect of the disclosed subject matter, and with reference to FIGS. **6A** and **6B** for purpose of illustration and not limitation, the hinge **321** is defined as a preset bend **681** in the lever **321**. The lever **321** can be made of a super elastic material, such as nitinol, which can bias the lever **321** toward the deployed condition **684**. If constructed using nitinol, the lever with hinge **324** (at bend **681**) can incur strains up to about 8%.

(38) With continued reference to FIGS. **6A** and **6B**, the actuation assembly **340** can include a slidable collar **682** disposed on the lever **321**. The slidable collar **682** can be formed of any suitable material, such as nitinol, Elgiloy, stainless steel, polyether block amide (PEBAX), silicone, or other metal or polymer. Metallic and polymeric combinations can also be incorporated to form the slidable collar in a braided tubular configuration. The slidable collar **682** can have a length spanning at least twice the hinge radius of curvature of the bend **681** with an overall length between 6 and 12 mm. Additional bends **681** can be incorporated to form a series to maintain the elasticity of the hinge **324** during a wider range of motion, as shown below in connection with FIGS. **8A** and

8B. In the case of incorporating multiple bends **681**, the slidable collar **682** can have a length that encompasses twice the hinge radius of curvature of all bends that are being straightened by the slidable collar. The actuation assembly **340** can also include a transmission member **685** having a rod, such as a hypotube, **686** adapted to slide the collar **682** between a delivery condition **683**, positioned over the hinge **324** to straighten the bend **681** of the lever, to a deployed condition **684**, with the bend **681** exposed. The collar **682** can be pushed or pulled through a transmission rod, such as a hypotube, **686** (i.e., catheter's hollowed elongate member), which acts to either position the collar **682** directly on the hinge **324** as shown in FIG. 6A for the delivery condition **683** or proximal to the hinge **324** to reveal and actuate the hinge as shown in FIG. 6B for the deployed condition **684**.

(39) In accordance with another aspect of the disclosed subject matter, the lever **321** can include a varied mechanical property along a length thereof selected from a varied thickness or a varied material property. For example, the lever **321** can include a stiffening member **690**, an exemplary embodiment of which is depicted in FIG. 6B, along a length thereof or can have a doubled-up portion to provide a varied thickness. Additionally or alternatively, the lever **321** can include a varied austenite finish temperature along its length. The varied mechanical property can ensure the lever applies a force of less than about 1.0 lbf to the native leaflet when the lever is transitioned to the deployed configuration.

(40) Embodiments of the disclosed subject matter with a slidable collar **682** can be used for valve repair, for example in edge-to-edge repair, in the same manner as described herein above. For example, in use, as the slidable collar **682** is moved proximally to expose the bend **681**, which rotates the coaptation aid **320** such that the atraumatic tip **325** is brought into contact with the distal (ventricular) side **104** of the anterior leaflet **103**. Further proximal movement of the slidable collar **682** further swings the lever **321** and atraumatic tip **325** upward until a fully deployed condition **684** is achieved. This rotation applies tension to the anterior leaflet **103** to bring it into a position located sufficiently close to the posterior leaflet **105** such that both leaflets can then be grasped and captured by arms of the leaflet fixation device **150**. The hinge radius of curvature shown in FIG. 6B is exemplary and a more or less dramatic curvature can be used.

(41) Once completed, the coaptation aid **320** can be removed by pushing collar **682** distally over the bend **681**, which will restore the slender profile of the delivery catheter for withdrawal from the body. The repair of the leaflets or tissue can be observed by non-invasive visualization techniques, such as echocardiography, to ensure the desired outcome. If the repair is not desired, the fixation device **150** can be repositioned or retrieved. Once the desired outcome is achieved, the fixation device **150** can then be detached from the distal end of the catheter shaft **111** and the catheter assembly **300** with the coaptation aid **320** can be withdrawn from the body.

(42) While a single bend **681** in lever **321** is illustrated in FIGS. 6A and 6B, the hinge **321** can have additional or alternative configurations. For example, as shown in FIGS. 7A and 7B, the bend **781** can include a coiled hinge. As shown in FIG. 7A, the coiled hinge **781** can be at least partially straightened and held under collar **682** in a delivery condition **683** during tracking and initial positioning prior to grasping the leaflets. As described above in connection with FIGS. 6A and 6B, slidable collar **682** can then be moved proximally to expose the hinge **781** as shown in FIG. 7B to move to a deployed condition **684** by activating the flexible coiled hinge and rotating the coaptation aid **320** such that the atraumatic tip **325** is brought into contact with the distal (ventricular) side **104** of the anterior leaflet **103**. While a single coil loop **781** is illustrated in FIGS. 7A and 7B, additional loops can be used to distribute strain more evenly and still ensure an adequate range of motion of the coaptation aid without exceeding about 8 percent strain in the, e.g., nitinol structure. The coiled hinge **781** can be nitinol or another shape memory material, and in some embodiments the preset shape can be tuned to overshoot closure of the lever beyond parallel to ensure reliable force application occurs for a grasped leaflet.

(43) FIGS. 8A and 8B illustrate another exemplary embodiment of bend of lever **321** in which the

bend **881** includes an series-style undulating hinge. As shown in FIG. **8A**, the undulating hinge **881** can be at least partially straightened and held under collar **682** in a delivery condition **683** during tracking and initial positioning prior to grasping the leaflets. As described above in connection with FIGS. **6A** and **6B**, slidable collar **682** can then be moved proximally to expose the hinge **881** as shown in FIG. **8B** to move to a deployed condition **684** by activating the flexible undulating hinge and rotating the coaptation aid **320** such that the atraumatic tip **325** is brought into contact with the distal (ventricular) side **104** of the anterior leaflet **103**. While a double undulating hinge **881** is illustrated in FIGS. **8A** and **8B**, additional undulations can be used to distribute strain more evenly and still ensure an adequate range of motion of the coaptation aid without exceeding about 8 percent strain in the, e.g., nitinol structure. The undulating hinge **881** can be nitinol or another shape memory material, and in some embodiments the preset shape can be tuned to overshoot closure of the lever beyond parallel to ensure reliable force application occurs for a grasped leaflet. (44) The lever can have any suitable dimensions, for example including a diameter of 0.5 to 2 mm and have a length that approximately equals 0.75 to 1.25 times the length of the implant. The atraumatic tip can have any suitable shape including spherical, cylindrical, or any other shape with rounded edges and can have any suitable dimensions including a diameter and length of approximately 3-5 mm and 6-9 mm. The atraumatic tip may be a wire form, coil, or hollow braided structure that can gently deform or ovalize to conform to tissue contacted during use. In this way, stresses induced in the leaflets are minimized. For example, FIGS. **9A**, **9B**, **9C**, **9D**, **9E**, and **9F** illustrate various configurations for atraumatic tip **325**. FIG. **9A** shows a cylindrical molded atraumatic tip **325**. FIG. **9B** shows an atraumatic tip **325** formed from a wireform **991**. In FIG. **9C**, the wireform **991** further includes flexing hinges **992** and **993**. FIG. **9D** illustrates a cylindrical molded atraumatic tip **325** having a covering **994**, e.g., formed of polyester, that can improve grip of the tip **325** on leaflets. FIG. **9E** illustrates an atraumatic tip **325** formed from a wireform **991** having a covering **995**, e.g., braided, that can that can improve grip of the tip **325** on leaflets. FIG. **9F** illustrates an atraumatic tip **325** formed from a wireform **991** further includes flexing hinges **992** and **993**, and having a two coverings **996** and **997**, e.g., textile, that can that can improve grip of the tip **325** on leaflets. While certain coverings have been described in connection with certain atraumatic tip configurations, it is readily apparent that alternative combinations of coverings and tip configurations can also be used.

(45) As embodied herein, coaptation aid **320** can be an optional accessory or permanently fixed to the catheter assembly **300**. For example, the coaptation aid **320** can be part of a kit (not shown) including a catheter shaft **111** and a leaflet fixation device **150**. Prior to a procedure, a user can observe the native anatomy and spacing between the leaflets to determine if a coaptation aid **320** will assist in the procedure. If so, the user can install or attach the coaptation aid **320** to the distal portion **313** of the catheter shaft of the delivery system. For example, the a proximal end **322** of lever of the coaptation aid **320** can be removably connected to the distal portion **313** of the catheter shaft, e.g., by snap fit or the like. Alternatively, the coaptation aid **320** can run alongside the catheter shaft **111** of the delivery system without direct attachment thereto.

(46) The embodiments illustrated herein are adapted for repair of a heart valve and have been described in connection with repair of a mitral valve, using an antegrade approach from a patient's left atrium. However, the disclosed subject matter can be used on any other leaflet that requires intraprocedural propping or positioning to make leaflet grasping and clipping easier. Other exemplary leaflets include, the posterior mitral leaflet, or any tricuspid, aortic, or pulmonary valve leaflet. Furthermore, while shown herein as a clip, the leaflet fixation device could be any suitable fixation means, including sutures, clips, staples, and the like. In addition, while a single device was shown and described in each embodiment herein, a plurality of devices or device features may be used to simultaneously support or stabilize two or more opposing leaflets in order to further simplify the edge-to-edge repair procedure. Further, while described in connection with an edge-to-edge repair procedure, the disclosure subject matter can be adapted to aid in other procedures, such

as leaflet resection, minimally invasive leaflet suturing, and chordae replacement.

(47) While the disclosed subject matter is described herein in terms of certain embodiments for purpose of illustration and not limitation, those skilled in the art will recognize that various modifications and improvements can be made to the disclosed subject matter without departing from the scope thereof. Moreover, although individual features of the disclosed subject matter may have been discussed herein or shown in the drawings of one embodiment and not in other embodiments, it is readily apparent that individual features of one embodiment can be combined with, or substituted for, one or more features of another embodiment or features from a plurality of embodiments.

(48) In addition to the specific embodiments claimed below, the disclosed subject matter is also directed to other embodiments having any other possible combination of the dependent features disclosed above, claimed below, and listed below:

(49) The actuation assembly can include a transmission member extending along a length of the catheter shaft to remotely transition the lever between the delivery configuration and the deployed configuration.

(50) The transmission member can include a pull wire to transition the lever from the delivery configuration to the deployed configuration by rotating the lever about the hinge.

(51) The lever can be biased toward the delivery configuration.

(52) The pull wire can extend through an alignment loop disposed on the distal portion of the catheter shaft.

(53) The pull wire can extend through a lumen along a length of the catheter shaft.

(54) The actuation assembly can further include a linkage coupled to the lever with the pull wire attached to the linkage.

(55) The linkage can be coupled to the lever between the hinge and the distal end.

(56) The lever is adapted to apply a force of less than about 1.0 lbf to the native leaflet when the lever is transitioned to the deployed configuration.

(57) The atraumatic tip can comprise rounded edges and can comprise one or more of nitinol, cobalt chromium, stainless steel, titanium, polyether block amide (PEBAX), polylactic acid (PLA), poly-L-lactide (PLLA), polylactic-co-glycolic acid (PLGA), polyurethane, polyethylene, polyester, and polyamide.

(58) The catheter assembly can further include a second lever coupled to the distal portion of the catheter shaft, the second lever having a proximal end, a distal end, and a hinge to transition the second lever between the delivery configuration and the deployed configuration.

(59) The second lever can further include a second atraumatic tip proximate the distal end of the second lever adapted to contact the distal side of the native leaflet.

(60) The actuation assembly can include a transmission member extending along a length of the catheter shaft to remotely transition the lever and the second lever simultaneously between the delivery configuration and the deployed configuration.

(61) The transmission member can include a pull wire attached to a linkage coupled to both the lever and the second lever to simultaneously transition the lever and the second lever from the delivery configuration to the deployed configuration by rotating each lever about its hinge.

(62) The pull wire can extend through an alignment loop disposed on the distal portion of the catheter shaft.

(63) The pull wire can extend through a lumen along a length of the catheter shaft.

(64) The hinge can be defined as a preset bend in the lever.

(65) The actuation assembly can include a slidable collar disposed on the lever.

(66) The collar can be slidable between a delivery condition positioned over the hinge to straighten the bend of the lever to the delivery configuration and a deployed condition with the bend exposed to allow the lever to transition to the deployed configuration.

(67) The lever can be made of a super elastic material.

- (68) The lever can be configured to incur a strain up to about 8% at the bend.
- (69) The actuation assembly can include a transmission member, which can be a rod adapted to slide the collar between the delivery condition and the deployed condition.
- (70) The rod can be a hypotube.
- (71) The lever can include a varied mechanical property along a length thereof to apply a force of less than about 1.0 lbf to the native leaflet when the lever is transitioned to the deployed configuration.
- (72) The varied mechanical property can be selected from a varied thickness or a varied material property.
- (73) The varied mechanical property can include a stiffening member along a length thereof.
- (74) The lever can be made of a material selected from the group consisting of a metal, a polymer, a composite, or a combination thereof.
- (75) The lever can include a wire.
- (76) The catheter assembly can further include a leaflet fixation device releasably coupled to the distal portion of the catheter shaft.
- (77) The leaflet fixation device can include a first gripper assembly and a second gripper assembly, each gripper assembly configured to capture a respective native leaflet.
- (78) As such, the particular features presented in the dependent claims and disclosed above can be combined with each other in other possible combinations. Thus, the foregoing description of specific embodiments of the disclosed subject matter has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosed subject matter to those embodiments disclosed.
- (79) It will be apparent to those skilled in the art that various modifications and variations can be made in the method and system of the disclosed subject matter without departing from the spirit or scope of the disclosed subject matter. Thus, it is intended that the disclosed subject matter include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A method for fixation of leaflets of a heart valve of a heart of a mammalian subject, comprising: advancing a leaflet fixation device coupled to a catheter assembly transvascularily to the heart valve; grasping a first leaflet of the heart valve via a first grasping assembly of the fixation device; actuating a coaptation aid from a delivery configuration to a deployed configuration such that an atraumatic tip of the coaptation aid contacts a distal side of a second leaflet of the heart valve and moves the second leaflet toward the first leaflet, the coaptation aid including a lever having a proximal end, a distal end comprising the atraumatic tip, and a hinge between the proximal and distal ends of the lever; and grasping the second leaflet of the heart valve via a second grasping assembly.
2. The method of claim 1, wherein the first leaflet is a posterior leaflet of the mitral valve, and the second leaflet is an anterior leaflet of the mitral valve.
3. The method of claim 2, wherein the advancing step includes introducing the catheter assembly in a femoral vein and advancing the leaflet fixation device across an interatrial septum of the heart to a mitral valve.
4. The method of claim 1, wherein the step of grasping the first leaflet includes: positioning a first arm of the first grasping assembly at a ventricular side of the first leaflet and a first gripping element at an atrial side of the first leaflet, and moving the first gripping element toward the first arm so as to capture the first leaflet therebetween.
5. The method of claim 1, wherein the step of grasping the first leaflet is performed prior to the actuating step.
6. The method of claim 1, wherein step of grasping the second leaflet is performed after the

actuating step and includes moving a second gripping element of the second grasping assembly toward a second arm assembly of the second grasping assembly to capture the second leaflet therebetween.

7. The method of claim 1, wherein the steps of grasping the first and second leaflets are performed simultaneously.

8. The method of claim 1, wherein, in the delivery configuration, the distal end of the lever is aligned with a catheter shaft of the catheter assembly, and, in the deployed configuration, the distal end of the lever extends transverse to the catheter shaft.

9. The method of claim 1, wherein the actuating step includes applying tension to a pull wire so as to pivot the distal end of the lever of the coaptation aid about the hinge, the pull wire being connected to a linkage between the distal end of the lever and the hinge.

10. The method of claim 9, wherein the pull wire extends through one of a lumen of a catheter shaft of the catheter assembly and an alignment loop located at a distal end of the catheter shaft.

11. The method of claim 1, wherein the hinge comprises a preset bend in the lever, and the actuating step includes sliding a slidable collar relative to the lever and the preset bend so as to expose the preset bend.

12. The method of claim 1, wherein the coaptation aid is coupled to a distal portion of a catheter shaft of the catheter assembly.

13. The method of claim 1, further comprising detaching the leaflet fixation device from a distal portion of a catheter shaft of the catheter assembly.

14. The method of claim 1, further comprising removing the coaptation aid from the heart of the mammalian subject.

15. The method of claim 14, further comprising transitioning the coaptation aid from the deployed configuration back to the delivery configuration prior to the removing step.

16. A method for fixation of leaflets of a heart valve of a heart of a mammalian subject, comprising: introducing a catheter assembly transvascularily into the mammalian subject so that a distal portion of a catheter shaft of the catheter assembly is positioned proximate the heart valve; deploying a first gripper assembly of a leaflet fixation device to capture a first leaflet of the heart valve, the leaflet fixation device being coupled to the distal portion of the catheter shaft; actuating a lever of a coaptation assembly of the catheter assembly from a delivery configuration toward a deployed configuration to position an atraumatic tip of the lever against a distal side of a second leaflet of the heart valve; tensioning the second leaflet toward the first leaflet by rotating a distal end of the lever about a hinge disposed between the distal end of the lever and a proximal end thereof, the distal end of the lever comprising the atraumatic tip; and deploying a second gripper assembly of the leaflet fixation device to capture the second leaflet.

17. The method of claim 16, wherein the steps of deploying the first gripper assembly and deploying the second gripper assembly are performed simultaneously.

18. The method of claim 16, further comprising removing the coaptation assembly from the mammalian subject.

19. A method for fixation of native leaflets of a heart valve, comprising: introducing a catheter assembly transvascularily, the catheter assembly comprising: an elongate catheter shaft having a proximal portion and a distal portion; a lever coupled to the distal portion of the catheter shaft, the lever having a proximal end, a distal end, and a hinge to transition the lever between an elongate delivery configuration generally aligned longitudinally with a distal portion of the catheter shaft and a deployed configuration extending transversely from the distal portion of the catheter shaft, the lever having an atraumatic tip proximal the distal end thereof; an actuation assembly coupled to the lever to remotely transition the lever between the delivery configuration and the deployed configuration; and a leaflet fixation device releasably coupled to the distal portion of the catheter shaft and comprising a first gripper assembly and a second gripper assembly; positioning the catheter assembly with the distal portion of the catheter shaft positioned proximate the heart valve;

deploying the first gripper assembly to capture a first native leaflet; actuating the actuation assembly to remotely transition the lever from the delivery configuration toward the deployed configuration to position the atraumatic tip against a distal side of a second native leaflet; and deploying the second gripper assembly to capture the second native leaflet.

20. The method of claim 19, wherein actuating the actuation assembly includes applying tension to a pull wire connected to the lever to rotate the distal end of the lever about the hinge from the delivery configuration to the deployed configuration, wherein rotating the distal end of the lever positions the atraumatic tip against the distal side of the second native leaflet to tension the second native leaflet toward the first native leaflet.
